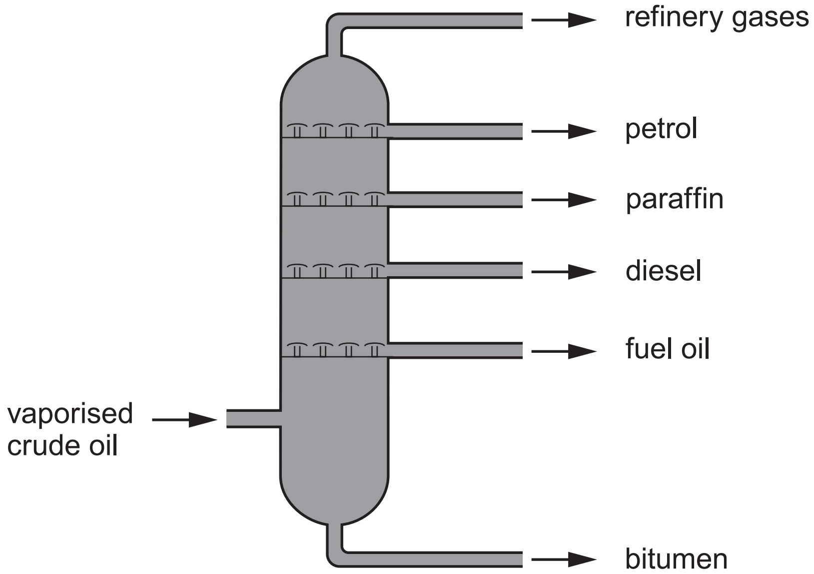
Examiner only

3. Crude oil is a mixture of hydrocarbon compounds. It is separated into different fractions inside a fractionating column. Each fraction has a different boiling point range.



(a) Explain how the length of the hydrocarbon chains within each fraction determines where each fraction collects in the column. [2]

....

....

....

(b) Many of the hydrocarbons that are collected from crude oil go through a second process called cracking. The equation shows the cracking of the hydrocarbon with the formula $C_{15}H_{32}$.

$$C_{15}H_{32} \longrightarrow C_3H_8 + 2C_2H_4 + 2 \text{ ....}$$

(i) **Complete the equation** for this reaction by adding the **formula** of **one** other product. [1]

(ii) Give the reaction conditions used in the process. [1]

....

(iii) Give **one** reason why it is important for oil companies to carry out cracking. [1]

....

....

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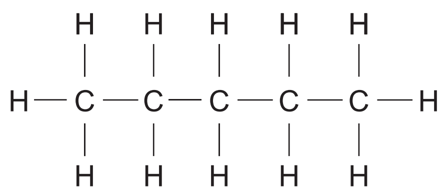
(c) Many hydrocarbons display isomerism.

(i) Give the meaning of the term *isomer*. [1]

....

....

(ii) C_5H_{12} has three isomers. The diagram shows one of these isomers.



Draw the **two** other isomers of C_5H_{12}. [2]

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(d) 8.4 g of a hydrocarbon was found to contain 7.2 g of carbon. Use this information to determine whether the compound is an alkane or an alkene. You **must** show your working. [3]

A_r(H) = 1 A_r(C) = 12

Conclusion ....

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